HKUST MATH 2431 - problem set 11

1.

1. $(*)$ **Convergence in probability** [4 Points]

Let $(\Omega, \mathscr{F}, \mathbf{P})$ a probability space. Suppose that for every $n \in \mathbb{N}$, we have random variables $X_n, Y_n : (\Omega, \mathscr{F}) \to (\mathbb{R}, \mathscr{B}(\mathbb{R}))$, and let $X, Y : (\Omega, \mathscr{F}) \to (\mathbb{R}, \mathscr{B}(\mathbb{R}))$ be random variables.

(a) Suppose that $X_n \xrightarrow[n\to\infty]{\mathbf{P}} X, Y_n \xrightarrow[n\to\infty]{\mathbf{P}} Y$. Show that

$$X_n + Y_n \xrightarrow[n\to\infty]{\mathbf{P}} X + Y.$$

1.a. For any $\varepsilon > 0$, we have

$\lim_{n\to\infty} P(|X_n - X| < \frac{\varepsilon}{2}) = 1$

Similarly for Y.

$$\{ |X_n - X| < \tfrac{\varepsilon}{2} \text{ and } |Y_n - Y| < \tfrac{\varepsilon}{2} \}$$
$$\subseteq \{ |(X_n + Y_n) - (X+Y)| < \varepsilon \}$$

$$P(|(X_n+Y_n) - (X+Y)| < \varepsilon)$$
$$\geq P(\{|X_n - X| < \tfrac{\varepsilon}{2}\} \cap \{|Y_n - Y| < \tfrac{\varepsilon}{2}\})$$
$$= \underbrace{P(|X_n - X| < \tfrac{\varepsilon}{2})}_{\to 1 \text{ as } n\to\infty} + \underbrace{P(|Y_n - Y| < \tfrac{\varepsilon}{2})}_{\to 1 \text{ as } n\to\infty} - \underbrace{P(\{|X_n - X| < \tfrac{\varepsilon}{2}\} \cup \{|Y_n - Y| < \tfrac{\varepsilon}{2}\})}_{\leq 1 \text{ (axioms of } P)}$$

$$\Rightarrow \liminf_{n\to\infty} P(|(X_n+Y_n) - (X+Y)| < \varepsilon) \geq 1$$

Coupled with $\limsup_{n\to\infty} P(|(X_n+Y_n) - (X+Y)| < \varepsilon) \leq 1$ by axioms of P,

finally $\lim_{n\to\infty} P(|(X_n+Y_n) - (X+Y)| < \varepsilon) = 1$

def of $X_n + Y_n \xrightarrow[n\to\infty]{P} X+Y$

(b) Let $(a_n)_{n\in\mathbb{N}}$ be a sequence of real numbers with $\lim_{n\to\infty} a_n = a$. Suppose that $X_n \xrightarrow[n\to\infty]{\mathbf{P}} x$, for some $x \in \mathbb{R}$. Show that

$$a_n \cdot X_n \xrightarrow[n\to\infty]{\mathbf{P}} a \cdot x.$$

1.b. For all large enough n, we can find $\varepsilon_1, \varepsilon_2$ s.t. for almost all $\omega \in \Omega$,

$|a_n - a| < \varepsilon_1 < 1$, $\quad |X_n(\omega) - x| < \varepsilon_2 < 1$

For any $|\varepsilon_1'| < \varepsilon_1$, $|\varepsilon_2'| < \varepsilon_2$

$$(a+\varepsilon_1')\cdot(x+\varepsilon_2')$$

$$= a\cdot x + \underbrace{a\cdot\varepsilon_2'}_{|\cdot| \le a\varepsilon_2} + \underbrace{\varepsilon_1'\cdot x}_{|\cdot| \le x\varepsilon_1} + \underbrace{\varepsilon_1'\cdot\varepsilon_2'}_{|\cdot| \le \varepsilon_1\varepsilon_2 < \max\{\varepsilon_1,\varepsilon_2\}}$$

Setting $\varepsilon = a\varepsilon_2 + x\varepsilon_1 + \varepsilon_1\varepsilon_2$

$$\le \underbrace{(2\max\{a,x\}+1)}_{\text{constant given } a,\, x;\ \text{does not depend on } n} \max\{\varepsilon_1,\varepsilon_2\}$$

implies $|a_n \cdot X_n(\omega) - a\cdot x| < \varepsilon$ for almost all $\omega \in \Omega$

This is equivalent to the def of $a_n \cdot X_n \xrightarrow[n\to\infty]{P} a\cdot x$.

4.

4. (∗) **Convergence in distribution** [4 Points]

Consider sequences of real random variables $(X_n)_{n\in\mathbb{N}}$ and $(Y_n)_{n\in\mathbb{N}}$ with probability density functions

(i) $f_{X_n}(x) = \lambda\left(1 - \frac{\lambda x}{n}\right)^{n-1} \mathbb{1}_{[0,\frac{n}{\lambda})}(x)$, where $\lambda > 0$,

(ii) $f_{Y_n}(x) = \frac{n+1}{n} x^{\frac{1}{n}} \mathbb{1}_{[0,1]}(x)$,

Show that X_n and Y_n converge in distribution as $n \to \infty$, and determine the respective limit distribution.

(i) $f_{X_n}(x) = \lambda\left(1 - \frac{\lambda x}{n}\right)^{n-1} \mathbb{1}_{[0,\frac{n}{\lambda})}(x)$, where $\lambda > 0$,

4. i. $\lim_{n\to\infty} f_{X_n}(x) = \lim_{n\to\infty} \lambda\left(1-\frac{\lambda x}{n}\right)^{n-1} \mathbb{1}_{[0,\frac{n}{\lambda})}(x)$

$$= \lambda \underbrace{\lim_{n\to\infty}\left(1-\frac{\lambda x}{n}\right)^{n}}_{\to e^{-\lambda x}} \frac{1}{1-\frac{\lambda x}{n}} \mathbb{1}_{[0,\frac{n}{\lambda})}(x)$$

$$= \lambda e^{-\lambda x} \frac{1}{1-\underbrace{\lim_{n\to\infty}\frac{\lambda x}{n}}_{\to 1}} \underbrace{\lim_{n\to\infty} \mathbb{1}_{[0,\frac{n}{\lambda})}(x)}_{\to \mathbb{1}_{[0,+\infty)}(x)}$$

$$= \lambda e^{-\lambda x} \mathbb{1}_{[0,+\infty)}(x)$$

So $X_n \xrightarrow[n\to\infty]{d} X \sim \text{Poisson}(\lambda)$.

(ii) $f_{Y_n}(x) = \frac{n+1}{n} x^{\frac{1}{n}} \mathbb{1}_{[0,1]}(x)$,

4. ii. $\lim_{n\to\infty} f_{Y_n}(x) = \lim_{n\to\infty} \frac{n+1}{n} x^{\frac{1}{n}} \mathbb{1}_{[0,1]}(x)$

$$= \underbrace{\left(\lim_{n\to\infty} \frac{n+1}{n}\right)}_{\to 1} \underbrace{\left(\lim_{n\to\infty} x^{\frac{1}{n}} \mathbb{1}_{[0,1]}(x)\right)}_{\to \mathbb{1}_{(0,1]}(x)}$$

$$= \mathbb{1}_{(0,1]}(x)$$

(for $x=0$, $x^{\frac{1}{n}} \to 0$ as $n\to\infty$)

So $Y_n \xrightarrow[n\to\infty]{d} Y \sim U((0,1])$.